
MultiVENT: Multilingual Videos of Events with Aligned Natural Text

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Abstract

Everyday news coverage has shifted from traditional broadcasts towards a wide range of presentation formats such as first-hand, unedited video footage. Datasets that reflect the diverse array of multimodal, multilingual news sources available online could be used to teach models to benefit from this shift, but existing news video datasets focus on traditional news broadcasts produced for English-speaking audiences. We address this limitation by constructing MultiVENT, a dataset of multilingual, event-centric videos grounded in text documents across five target languages. MultiVENT includes both news broadcast videos and non-professional event footage, which we use to analyze the state of online news videos and how they can be leveraged to build robust, factually accurate models. Finally, we provide a model for complex, multilingual video retrieval to serve as a baseline for information retrieval using MultiVENT.

1 Introduction

Information dissemination for current events has traditionally consisted of professionally collected and produced materials, leading to large collections of well-written news articles and high-quality videos. As a result, such materials form the basis for significant prior work in content analysis and retrieval [52, 20, 2, 15, 48]. Meanwhile, a high volume of event-centric content today is generated by non-professionals, such as on-the-scene witnesses to events who hastily capture videos and upload them to the internet without further editing. We propose that this contemporary landscape of news content can be leveraged by models to produce a more comprehensive understanding of events. News agencies have adapted to this shift, often collecting and incorporating this online content into official broadcasts, but news video datasets still do not typically address this new domain of event coverage.

In addition to focusing on traditional news sources, existing news video datasets predominantly consider content produced in English. This is consistent with common practices in video dataset collection: Collected videos and captions are recorded in English, and when multilinguality is considered, it is achieved by directly translating captions and transcripts [46, 23, 38, 19]. Because this data is originally produced for English speaking audiences, these multilingual datasets can contain unwanted content biases like "translationese" [6, 21]. As event-centric video content produced in other languages makes up a large portion of news videos online, we argue that including organic, multilingual content is necessary for a diverse and perspective-agnostic sampling of event coverage.

With these ideas in mind, we present MultiVENT, a dataset of **Multilingual Videos of Events** with aligned **Natural Text** that contains 2,396 diverse, event-centric videos and text descriptions

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Figure 1: A sample video-text pair from MultiVENT. Every event-centric video is paired with a corresponding video description and a long-form text document describing the event, both in the same language as the video. If the language is not English, the video is also paired with a corresponding English document.

that reflect the distribution of news content online. The videos are grounded in natural language video descriptions and long-form text documents, and the data spans 260 current events across over forty countries. The content in MultiVENT is collected in five target languages: Arabic, Chinese, English, Korean, and Russian, and as the multilinguality is organic, the data is less likely to suffer from translation bias. We provide an illustration of the dataset’s contents in Figure 1: Each natural language query (describing a video of a current event) is paired with grounding text documents and a unique corresponding video. We use MultiVENT to explore and characterize the variety of event-centric videos available online and illustrate the importance of leveraging these different video types when building multimodal information systems.

Citizen journalism, the most notable example being Wikipedia [18], has emerged alongside other online news sources as a method for curating comprehensive summaries of events. Work in natural language processing has considered the problem of automating this process by training models to generate informative reports using online source materials [25, 43, 32]. We use MultiVENT to explore how this process can be extended to incorporate multimodal sources of evidence. As a first step in this direction, we consider the task of video retrieval on MultiVENT, through which a model learns to retrieve multimodal source material given a natural language event description. This task differs from prior video retrieval benchmarks [9, 51, 1, 28, 49] as the videos in MultiVENT vary widely in length and content presentation, are multilingual, and can involve significant amounts of on-screen text. In addition to multilingual natural language captions for each video, we provide full text documents that ground the events and serve as more complex retrieval queries.

In summary, our contributions are:

1. We present MultiVENT, a multimodal, multilingual information retrieval dataset of grounded videos depicting current events. The dataset targets five languages and covers a range of online video formats beyond traditional news broadcasts.
2. Using MultiVENT, we quantitatively illustrate the information presented by news videos and the differences in content between video formats, and qualitatively evaluate how multimodal coverage of an event can evolve over time.
3. We present MultiCLIP, a model for multilingual, event-centric video retrieval that serves as a baseline for video retrieval approaches on the task.

2 Related Work

2.1 Video retrieval datasets

Early video datasets generally contained short clips spanning narrow ranges of topics, such as the Microsoft Research Video Description Corpus [9]. Video datasets spanning larger domains include

MSR-VTT [51] and DiDeMo [1], although the lengths of these videos were still relatively short. The V3C dataset [37, 3] offered longer video lengths while still spanning a wide range of topics such as news reports. A shift towards massive video datasets was instigated by HowTo100M [28], which included over 130 million video clips belonging to one million narrated instructional videos. VaTeX [46], released in the same year, considered video retrieval from a multilingual context using caption translation. Additional multilingual video retrieval datasets include Rudder [13], consisting of instructional videos for making toys with multilingual captions, MTRV [23], which extended the TVR dataset [24] by adding Chinese subtitles and queries, and Multi-HowTo100M [19], which extended HowTo100M by scraping YouTube for subtitles in up to 9 other languages. Recently, Chen et al. [8] released the ChinaOpen dataset which contains a wide range of video-caption pairs originally produced in Chinese. Recent work has also considered the problem of interpreting text-heavy video content: Wu et al. [49] and Jahagirdar et al. [20] introduced datasets that focus on within-video text and OCR annotations, including news broadcasts.

2.2 Video retrieval methods

The size of early video datasets allowed retrieval systems to rely on pre-extracted features from expert systems like action recognition models. As massive video datasets gained prominence, the video retrieval paradigm moved towards ad-hoc video-text feature extraction using large pretrained models. Dosovitskiy et al. [14] proposed using stand-alone transformer architectures for video understanding, and Bertasius et al. [7] showed that applying space- and time-based self-attention independently improved performance. Bain et al. applied findings directly to video retrieval, training and evaluating transformer architectures on WebVid-2M [4]. Radford et al. [33] introduced CLIP and showed that pretraining models to match captions to images can result in scalable models, and CLIP’s applicability to video retrieval was demonstrated by Fang et al. [16] through their CLIP2Video model. More fine-grained modifications to CLIP were proposed. Wang et al. [45] introduced "Object-aware Transformers", which extended video-text transformers to incorporate object-level annotations within video footage, and Ge et al. [17] modified the pretraining task to involve teaching a vision-text model to answer multiple choice questions about a video. Bain et al. [5] adapted large image-text models to the task of long video retrieval by incorporating the weighted-mean of frame embeddings, and Wu et al. [49] incorporated independent optical character recognition and embeddings into the encoder pipeline to explicitly model in-video text.

2.3 Report generation using online sources

A wide range of research has used online corpora for report generation tasks, including QA-pair and knowledge graph generation [35, 34, 53, 29, 22, 30]. Notably, Lewis et al. [25] introduced a method for automatically extracting question-answer pairs from large corpora of text documents, and applied this method to Wikipedia to produce the PAQ dataset. Some PAQ extensions have been multilingual — Pisare et al. [31] built the WikiOmnia QA dataset on Russian Wikipedia documents, and Rybak et al. [40] produced a question-Wikipedia passage dataset in Polish. Recently, Qian et al. [32] extended the ideas in PAQ to construct WebBrain, a task in which a model must generate factual articles with references given a natural language query. In the multimodal domain, Reddy et al. and Chen et al. have considered the problem of open-domain QA for image-text data [36, 10], with Chen et al. using Wikipedia to generate a multimodal dataset. In a similar vein, Li et al. propose a dataset for information extraction from multimedia articles [27] and an extraction approach that can be used with text, image, and video content [26].

3 Dataset

In this section we outline the MultiVENT collection process. The dataset includes 2,396 videos and corresponding text descriptions covering 260 current events grounded in 468 text documents, and includes content in Arabic, Chinese, English, Korean, and Russian. We first identify 260 visually salient current events spanning from 2013 to 2023, and assign a target language to each event. Then, for each event, we collect grounding text documents and a set of videos in the event’s target language.